

Derivation of the Ultrasound RBC Sensor SNR

Grand Unified Theory of Imaging (GUTI)

Self-contained note from repository assumptions and result metadata.

Summary

The ultrasound summary table reports an amplitude SNR of approximately 0.1445. This is not the SNR of a single isolated source-sensor pair. It is the RMS output amplitude of the full cone-scaled RBC pressure operator, restricted to one 2 MHz axial range slice and multiplied by a 1% CBV-variability factor, divided by the assumed scalar sensor noise.

$$\text{SNR}_{\text{sensor}} = \frac{0.00072268 \text{ Pa}}{0.005 \text{ Pa}} = 0.1445$$

Assumptions and References

The numerical assumptions are taken from these local repository references:

- RBC scaling and result metadata: [guti/modalities/us/analytical.py](#).
- Summary constants and SNR calculation: [scripts/make_modality_noise_summary.py](#).
- 6000-sensor cone-slice 1% CBV JSON result: [results/variants/us_free_field_analytical_50khz_128k_rbc_cone_slice_cbv1pct_20260602/json/001_50khz_128000src_6000sensors.json](#).
- Generated summary explanation: [results/modality_correlated_noise_summary/README.md](#).

$$f_{\text{RBC}} = 2 \text{ MHz}, \quad f_{\text{ref}} = 50 \text{ kHz}, \quad P_{\text{external}} = 1 \text{ MPa}$$

$$\text{BSC}_{10\text{MHz}} = 3 \times 10^{-5} \text{ cm}^{-1}\text{sr}^{-1}, \quad \text{CBV} = 0.03$$

$$V_{\text{voxel}} = 24 \text{ mm}^3, \quad r_{\text{ref}} = 0.10 \text{ m}, \quad T_{\text{in}} = 0.5, \quad T_{\text{out}} = 0.1$$

Important: the 24 mm^3 voxel volume is the backscatter/source-cell volume used in the 50 kHz reference simulation basis. It is not a claim that the 2 MHz image-resolution voxel has volume 24 mm^3 . The later 2 MHz extrapolation uses the explicit spatial-mode scaling below.

$$\left(\frac{f_{\text{RBC}}}{f_{\text{ref}}}\right)^3 = \left(\frac{2 \text{ MHz}}{50 \text{ kHz}}\right)^3 = 64,000$$

RBC Backscatter Model

The blood backscatter coefficient is scaled from the 10 MHz reference value using the f^4 dependence used by the run:

$$\text{BSC}_{\text{blood}}(f) = \text{BSC}_{10\text{MHz}} \left(\frac{f}{10 \text{ MHz}}\right)^4$$

$$\text{BSC}_{\text{blood}}(2 \text{ MHz}) = 3 \times 10^{-5} \left(\frac{2}{10}\right)^4 = 4.8 \times 10^{-8} \text{ cm}^{-1}\text{sr}^{-1}$$

Multiplying by cerebral blood volume gives:

$$\eta = \text{CBV} \cdot \text{BSC}_{\text{blood}} = 0.03 \cdot 4.8 \times 10^{-8}$$

$$\eta = 1.44 \times 10^{-9} \text{ cm}^{-1}\text{sr}^{-1} = 1.44 \times 10^{-7} \text{ m}^{-1}\text{sr}^{-1}$$

The pairwise pressure model is:

$$p_{ij} = P_{external} T_i T_j \frac{\sqrt{\eta V_{voxel}}}{r_{ij}}$$

The square root appears because the BSC gives an intensity-like scattering fraction, while the channel model uses pressure amplitude.

Reference Pair Pressures

$$\frac{\sqrt{\eta V_{voxel}}}{r} = \frac{\sqrt{(1.44 \times 10^{-7})(24 \times 10^{-9})}}{0.10} = 5.8788 \times 10^{-7}$$

$$p_{no\ skull} = 10^6 \cdot 5.8788 \times 10^{-7} = 0.58788\text{ Pa}$$

Case	Formula	Pressure
outside/outside	0.58788 x 0.1 x 0.1	0.0058788 Pa = 5.8788 mPa
inside/outside	0.58788 x 0.5 x 0.1	0.029394 Pa = 29.394 mPa
inside/inside	0.58788 x 0.5 x 0.5	0.14697 Pa = 146.97 mPa

These are baseline 10 cm pair pressures before the 1% CBV-variability factor. The variable component used in the summary table is 0.01 times the pressure response.

From Pair Pressures to Table RMS

The table uses the full cone-scaled operator, not one reference pair. The stored 6000-sensor SLQ metadata reports:

$$\|G\|_F^2 = 372368.726, \quad n_{outputs} = 366000$$

$$A_{all} = \sqrt{\frac{\|G\|_F^2}{n_{outputs}}} = \sqrt{\frac{372368.726}{366000}} = 1.00866\text{ Pa}$$

The row then restricts source power to one axial range slice for a two-cycle, 2 MHz pulse:

$$f_{slice} = \frac{c(2/f_{RBC})/2}{D} = \frac{1540(2/2,000,000)/2}{0.15} = 0.00513333$$

The displayed table amplitude is:

$$A_{table} = A_{all} \sqrt{f_{slice}} \cdot 0.01$$

$$A_{table} = 1.00866 \sqrt{0.00513333} \cdot 0.01 = 0.00072268\text{ Pa} = 0.72268\text{ mPa}$$

Sensor Noise and Final SNR

The summary row assumes scalar iid pressure noise:

$$\sigma_{sensor} = 5\text{ mPa} = 0.005\text{ Pa}$$

$$SNR_{sensor} = \frac{A_{table}}{\sigma_{sensor}} = \frac{0.00072268}{0.005} = 0.1445359$$

Interpretation

The number 0.1445 is a table-level RMS amplitude SNR under the current RBC-cone assumptions. It combines BSC-derived RBC pressure scaling at 2 MHz, cone-dependent skull transmission, the 50 kHz reference simulation basis using $V = 24 \text{ mm}^3$, restriction to one two-cycle 2 MHz axial slice, a 1% CBV-variability amplitude factor, and fixed 5 mPa scalar sensor noise. It should not be read as the SNR of every individual voxel-to-sensor pair.

Repository References

- [guti/modalities/us/analytical.py](#)
- [scripts/make_modality_noise_summary.py](#)
- [results/variants/us_free_field_analytical_50khz_128k_rbc_cone_slice_cbv1pct_20260602/json/001_50khz_128000src_6000sensors.json](#)
- [results/modality_correlated_noise_summary/README.md](#)